Diogo M. Camacho

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Executive Profile

Transformative computational AI leader with 15+ years of experience building and deploying AI platforms at the intersection of biology and drug discovery, across pharma, venture-backed startups, and translational research institutes.

- **Enterprise AI Strategy**: Build and lead computational AI organizations that drive portfolio differentiation, shape corporate strategy, and create option value in novel drug discovery platforms; Leading AI transformations to accelerate drug discovery.
- **Technical Innovation**: Deliver proprietary AI platforms, patient stratification algorithms, active learning frameworks, and agentic AI systems across the full drug discovery lifecycle, from target identification through preclinical validation.
- **Team Building & Fundraising**: Scaling teams from first hire to cross-disciplinary groups of 25+; deep track record in mentorship, succession planning, and recruiting computational science leaders across the industry; organizin data rooms for venture funding.
- **Executive Presence & Credibility**: Board and VC interface across multiple companies; secured DARPA and NIH funding; 20+ peer-reviewed publications (7,000+ citations) and 6 patents spanning computational biology, ML, and synthetic biology.

Professional Experience

Flagship Pioneering
Cambridge, MA

Senior Principal, Computation — *Oct 2024 - Present*

- **Company origination**: AI leader on founding team of FL111, leading the computational efforts from exploration to seed stage funding; Advisory roles on FL106, FL108, FL115, from exploration to seed stage.
- **Venture platform strategy**: Computational and data strategy for multiple seed-stage ventures (FL110, FL111, FL121), directly developing platform capabilities and setting the technical direction for early-stage programs.
- **Agentic AI systems**: Pioneering agentic AI approaches for drug discovery and direct-to-consumer health applications across the Flagship portfolio.
- **Talent and leadership**: Recruiting, training, mentoring, and retaining computational science leaders across multiple portfolio companies (Ampersand, FL121, FL122, FL110, FL115).

Abiologics *(Flagship Pioneering portfolio company)*
Cambridge, MA

Head of Computational Sciences — *Jan 2025 - Present*

- **Platform strategy**: Scaling and leading the computational sciences function for a D-amino acid therapeutic platform through AI-first design: computational and AI strategy, operations, and execution.
- **Active learning**: Established active learning as the core experimental design paradigm for therapeutic mini-binder optimization, enabling data-driven prioritization across the DBT cycle.
- **Agentic AI**: Architected and deployed agentic AI systems for target identification, target prioritization, and competitive intelligence, reducing manual analysis burden and accelerating program decisions.
- **Organizational leadership**: Own all aspects of team growth and function: hiring, performance management, budgeting, and executive-level search.

Cellarity
Boston, MA

Vice President, Computational and Data Sciences — *Jul 2022 - Feb 2024*

- **Executive leadership**: Member of R&D Leadership Team reporting to the CSO; shaped corporate strategy at the intersection of computational platform and therapeutic portfolio direction.
- **Organizational build**: Scaled and led a 25-person computational organization across bioinformatics, ML, and computational biology, with teams in Boston and Germany.
- **Platform strategy**: Directed end-to-end platform strategy and operations, integrating Cellarity's computational capabilities into R&D programs and investor narratives that shaped fundraising outcomes.
- **External engagement**: Primary computational interface with board members, founding VC partners, and external collaborators on platform differentiation and roadmap.

Rheos Medicines
Boston, MA

SENIOR DIRECTOR, COMPUTATIONAL BIOLOGY AND BIOINFORMATICS
Nov 2020 - Jul 2022

- **Precision medicine strategy**: Established the computational precision medicine strategy for immunometabolism drug discovery, defining how patient stratification and multi-omics data would inform program decisions.
- **AI and data strategy**: Defined and implemented Rheos' full AI and data strategy in partnership with Research Informatics, ensuring integration across discovery and translational programs.
- **Platform delivery**: Delivered proprietary patient stratification algorithms and multi-omics platform adopted across the pipeline.
- **Board engagement**: Presented computational strategy to board and pharma partners, directly supporting business development and partnership decisions.

Wyss Institute @ Harvard University
Boston, MA

LEAD, PREDICTIVE BIOANALYTICS
Jul 2016 - Nov 2020

- **Founding leadership**: Founded and led the Predictive BioAnalytics Initiative — a first-of-kind ML/AI function embedded in translational science — defining research strategy, securing funding, and building the team from scratch.
- **Capital acquisition**: Raised >$2.4M in competitive federal grants through DARPA and NIH, positioning the initiative as an externally validated platform for AI-driven translational discovery.
- **Infectious disease research**: Co-led computational contributions to host immune tolerance research, using computational transcriptomics to identify infection response pathways validated experimentally — published in Advanced Science.
- **Team leadership**: Directed and mentored interdisciplinary teams of staff scientists, postdocs, graduate students, and interns across synthetic biology, systems biology, and ML.

Evelo Biosciences
Cambridge, MA

SENIOR SCIENTIST, COMPUTATIONAL SYSTEMS BIOLOGY LEAD
Jan 2015 - Apr 2016

Ember Therapeutics
Cambridge, MA

PRINCIPAL SCIENTIST, COMPUTATIONAL SYSTEMS BIOLOGY LEAD
Jan 2014 - Dec 2014

Pfizer, Inc.
Cambridge, MA

SENIOR RESEARCH SCIENTIST, COMPUTATIONAL SCIENCES CENTER OF EMPHASIS
Jan 2011 - Jan 2014

Howard Hughes Medical Institute @ Boston University
Boston, MA

POSTDOCTORAL FELLOW (JIM COLLINS LAB)
Jul 2007 - Jan 2011

Education

Virginia Polytechnic Institute and State University
Blacksburg, VA

PH.D. IN GENETICS, BIOINFORMATICS, AND COMPUTATIONAL BIOLOGY

Faculdade de Ciencias da Universidade de Lisboa
Lisboa, Portugal

B.SC. IN BIOCHEMISTRY

Publications and Patents

- Marbach D, Costello JC, Kuffner R, Vega NM, Prill RJ, **Camacho DM**, Allison KR, Kellis M, Collins JJ, Stolovitzky G (2012). Wisdom of crowds for robust gene network inference. *Nature Methods* 9, 796-804.
- **Camacho DM**, Collins KM, Powers RK, Costello JC, Collins JJ (2018). Next-generation machine learning for biological networks. *Cell* 173, 1581-1592.
- Valeri J, Collins KM, Ramesh P, Alcantar M, Lepe BA, Lu TK, **Camacho DM** (2020). Sequence-to-function deep learning frameworks for engineered riboregulators. *Nature Communications* 11, 5058.
- Bojar D, Powers RK, **Camacho DM**, Collins JJ (2020). Deep-learning resources for studying glycan-mediated host-microbe interactions. *Cell Host & Microbe* 29, 132-144.
- Gazzaniga FS, **Camacho DM**, Wu M, Palazzo MFS, Dinis ALM, Grafton FN, Cartwright MJ, Super M, Kasper DL, Ingber DE (2021). Harnessing colon chip technology to identify commensal bacteria that promote host tolerance to infection. *Frontiers in Cellular and Infection Microbiology* 11, 638014.
- Sperry MM, Novak R, Keshari V, Dinis ALM, Cartwright MJ, **Camacho DM**, Pare JF, Super M, Levin M, Ingber DE (2022). Enhancers of host immune tolerance to bacterial infection discovered using linked computational and experimental approaches. *Advanced Science* 9, 2200222.
- Valeri JA, Soenksen LR, Collins KM, Ramesh P, Cai G, Powers R, **Camacho DM**, et al. (2023). BioAutoMATED: An end-to-end automated machine learning tool for explanation and design of biological sequences. *Cell*

Systems 14, 525.
- Camacho D, de la Fuente A, Mendes P (2005). The origin of correlations in metabolomics data. *Metabolomics* 1, 53-63.

Patents spanning drug discovery, ML methods, and synthetic biology.

Full publication list available on Google Scholar (6,770+ citations).

Grants

Raised >$2.4M in competitive federal funding (DARPA, NIH) — Competitive federal funding to support translational bioanalytics platforms.